Srednicki QFT: Chapter 34

Left- and Right-Handed Spinor Fields

Cadabra2 expressions and numerical verification

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Metric Convention Summary

All formulas in this document are derived in **both** the West coast and East coast metric conventions. The table below summarises the key differences:

Property	West coast $(+---)$	East coast $(-+++)$
$g^{\mu\nu}$	$\mathrm{diag}(+1,-1,-1,-1)$	$\mathrm{diag}(-1,+1,+1,+1)$
g^{00}	$+1$	-1
g^{11}	-1	$+1$
Signature	$+2$	-2
Used by	Srednicki, Weinberg	Peskin–Schroeder, most particle physicists

Key consequence: any formula involving $g^{\mu\nu}$ (boosted generators, σ^μ, trace identities) differs by signs between conventions. Formulas involving only spatial indices (g^{ij}) pick up a sign flip (-1 in WC vs. $+1$ in EC), while the time component g^{00} flips ($+1$ in WC vs. -1 in EC).

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1 The Lorentz Algebra and its $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ Structure

1.1 The six generators of $\mathrm{SO}(1,3)$

The Lorentz group $\mathrm{SO}(1,3)$ is a six-dimensional Lie group. Its Lie algebra $\mathfrak{so}(1,3)$ is spanned by six generators, conventionally split into:

- **Rotations:** $J_i = \frac{1}{2}\varepsilon_{ijk}M^{jk}, \quad i = 1,2,3$
- **Boosts:** $K_i = M^{i0}, \quad i = 1,2,3$

where $M^{\mu\nu} = -M^{\nu\mu}$ are the antisymmetric generators of the defining (4-dimensional vector) representation.

The commutation relations of the Lorentz algebra are:

$$[J_i,\, J_j] = i\,\varepsilon_{ijk}\,J_k \tag{1}$$
$$[J_i,\, K_j] = i\,\varepsilon_{ijk}\,K_k \tag{2}$$
$$[K_i,\, K_j] = -i\,\varepsilon_{ijk}\,J_k \tag{3}$$

Equation (1) says the rotations by themselves close into an $\mathfrak{su}(2)$ algebra. Equation (2) says the boosts K_i transform as a 3-vector under rotations — exactly what one expects from a physical boost parameter. Equation (3) is the key relation: two boosts do *not* close into another boost; instead they produce a rotation (Thomas precession). The crucial minus sign in (3) is a direct consequence of the Lorentzian signature: in SO(4) (Euclidean) one would get $+i\varepsilon_{ijk}J_k$.

Remark. Equations (1)–(3) are convention-independent: they hold in both the West coast $(+---)$ and East coast $(-+++)$ signatures when J_i and K_i are defined as above. (One can verify this from the master commutator $[M^{\mu\nu}, M^{\rho\sigma}] = i(g^{\nu\rho}M^{\mu\sigma} - g^{\mu\rho}M^{\nu\sigma} - g^{\nu\sigma}M^{\mu\rho} + g^{\mu\sigma}M^{\nu\rho})$ by substituting the explicit definitions of J_i and K_i.)

1.2 The N_i, $N_i^\dagger$ combinations

Define new combinations (Srednicki eq. 33.16):

$$N_i \equiv \tfrac{1}{2}(J_i - iK_i), \tag{4}$$
$$N_i^\dagger \equiv \tfrac{1}{2}(J_i + iK_i). \tag{5}$$

We claim these satisfy

$$[N_i, N_j] = i\varepsilon_{ijk}N_k, \tag{6}$$
$$[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger, \tag{7}$$
$$[N_i, N_j^\dagger] = 0. \tag{8}$$

We now derive each relation explicitly, step by step.

1.3 Derivation of $[N_i, N_j] = i\varepsilon_{ijk}N_k$

Step 1: Expand the commutator.

$$\begin{aligned}[N_i, N_j] &= \Big[\tfrac{1}{2}(J_i - iK_i),\ \tfrac{1}{2}(J_j - iK_j)\Big] \\ &= \tfrac{1}{4}\Big[J_i - iK_i,\ J_j - iK_j\Big]. \end{aligned} \tag{9}$$

Using bilinearity of the commutator:

$$\begin{aligned}[J_i - iK_i,\ J_j - iK_j] &= [J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] + i^2[K_i, K_j] \\ &= [J_i, J_j] - i[J_i, K_j] - i[K_i, J_j] - [K_i, K_j]. \end{aligned} \tag{10}$$

Step 2: Substitute the three Lorentz commutators.

$$\begin{aligned}[J_i, J_j] &= i\varepsilon_{ijk}J_k, \\ [J_i, K_j] &= i\varepsilon_{ijk}K_k, \\ [K_i, J_j] &= -[J_j, K_i] = -i\varepsilon_{jik}K_k = +i\varepsilon_{ijk}K_k, \\ [K_i, K_j] &= -i\varepsilon_{ijk}J_k. \end{aligned}$$

(In the third line we used $\varepsilon_{jik} = -\varepsilon_{ijk}$.)

Step 3: Substitute into (10)**.**

$$\begin{aligned}[J_i - iK_i,\ J_j - iK_j] &= i\varepsilon_{ijk}J_k - i(i\varepsilon_{ijk}K_k) - i(i\varepsilon_{ijk}K_k) - (-i\varepsilon_{ijk}J_k) \\ &= i\varepsilon_{ijk}J_k + \varepsilon_{ijk}K_k + \varepsilon_{ijk}K_k + i\varepsilon_{ijk}J_k \\ &= 2i\varepsilon_{ijk}J_k + 2\varepsilon_{ijk}K_k \\ &= 2\varepsilon_{ijk}(iJ_k + K_k). \end{aligned} \tag{11}$$

Step 4: Identify the result. Note that $iJ_k + K_k = i(J_k - iK_k)$. Therefore:

$$[J_i - iK_i,\ J_j - iK_j] = 2\varepsilon_{ijk} \cdot i(J_k - iK_k) = 4i\varepsilon_{ijk} \cdot \tfrac{1}{2}(J_k - iK_k) = 4i\varepsilon_{ijk}N_k. \tag{12}$$

Multiplying by $\frac{1}{4}$:

$$\boxed{[N_i, N_j] = \tfrac{1}{4} \cdot 4i\varepsilon_{ijk}N_k = i\varepsilon_{ijk}N_k.} \tag{13}$$

Key insight on signs. The crucial role of the sign in $[K_i, K_j] = -i\varepsilon_{ijk}J_k$ is visible in Step 3: the $i^2 = -1$ from the $(-i)(-i)$ prefactor *combined* with the minus sign in $[K_i, K_j]$ produces a *double negative*, giving $+i\varepsilon_{ijk}J_k$ contribution that adds constructively with the $[J_i, J_j]$ term. Had we instead had $[K_i, K_j] = +i\varepsilon_{ijk}J_k$ (as in $\mathfrak{so}(4)$), the J terms would cancel and we would not obtain a closed $\mathfrak{su}(2)$ algebra for N_i.

1.4 Derivation of $[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk} N_k^\dagger$

By an identical calculation with $iK \to -iK$:

$$\begin{aligned}
[J_i + iK_i,\ J_j + iK_j] &= [J_i, J_j] + i[J_i, K_j] + i[K_i, J_j] + i^2[K_i, K_j] \\
&= i\varepsilon_{ijk} J_k + i(i\varepsilon_{ijk} K_k) + i(i\varepsilon_{ijk} K_k) + (-1)(-i\varepsilon_{ijk} J_k) \\
&= i\varepsilon_{ijk} J_k - \varepsilon_{ijk} K_k - \varepsilon_{ijk} K_k + i\varepsilon_{ijk} J_k \\
&= 2i\varepsilon_{ijk} J_k - 2\varepsilon_{ijk} K_k \\
&= 2\varepsilon_{ijk}(iJ_k - K_k) = 4i\varepsilon_{ijk} \cdot \tfrac{1}{2}(J_k + iK_k) = 4i\varepsilon_{ijk} N_k^\dagger. \qquad (14)
\end{aligned}$$

Hence:

$$\boxed{[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk} N_k^\dagger.} \qquad (15)$$

1.5 Derivation of $[N_i, N_j^\dagger] = 0$

$$\begin{aligned}
[N_i, N_j^\dagger] &= \Big[\tfrac{1}{2}(J_i - iK_i),\ \tfrac{1}{2}(J_j + iK_j)\Big] \\
&= \tfrac{1}{4}\Big[J_i - iK_i,\ J_j + iK_j\Big] \\
&= \tfrac{1}{4}\Big([J_i, J_j] + i[J_i, K_j] - i[K_i, J_j] - i^2[K_i, K_j]\Big) \\
&= \tfrac{1}{4}\Big(i\varepsilon_{ijk} J_k + i(i\varepsilon_{ijk} K_k) - i(i\varepsilon_{ijk} K_k) + [K_i, K_j]\Big) \\
&= \tfrac{1}{4}\Big(i\varepsilon_{ijk} J_k - \varepsilon_{ijk} K_k + \varepsilon_{ijk} K_k + (-i\varepsilon_{ijk} J_k)\Big) \\
&= \tfrac{1}{4}\Big(i\varepsilon_{ijk} J_k - i\varepsilon_{ijk} J_k\Big) = 0. \qquad (16)
\end{aligned}$$

$$\boxed{[N_i, N_j^\dagger] = 0.} \qquad (17)$$

1.6 The isomorphism statement — over $\mathbb{C}$

The computation above shows that the six operators $\{N_i, N_i^\dagger\}$ split into two commuting copies of the $\mathfrak{su}(2)$ algebra. However, one must be careful about the field:

Remark (Over $\mathbb{R}$: no isomorphism). Over the reals, $\mathfrak{so}(1,3) \ncong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$. The reason is that the generators K_i of boosts are *not Hermitian* (and cannot be made so while keeping J_i Hermitian, because the Lorentz group is non-compact). Consequently $N_i^\dagger$ is the Hermitian conjugate of N_i, not an independent generator: there are only 6 real independent generators, not 6+6. The real form $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ has only *compact* unitary representations, while $\mathrm{SO}(1,3)$ has no finite-dimensional unitary representations (other than the trivial one).

Remark (Over $\mathbb{C}$: the isomorphism holds). Consider the *complexification* $\mathfrak{so}(1,3)_{\mathbb{C}} \equiv \mathfrak{so}(1,3) \otimes_{\mathbb{R}} \mathbb{C}$. This is the complex Lie algebra obtained by allowing complex linear combinations of the real generators $\{J_i, K_i\}$. In this larger algebra, N_i and $N_i^\dagger$ *are* independent (not related by a reality condition), and our calculation proves:

$$\mathfrak{so}(1,3)_{\mathbb{C}} \;\cong\; \mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C}) \;\cong\; \mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}}. \tag{18}$$

The two copies are called $\mathfrak{su}(2)_L$ (generated by N_i) and $\mathfrak{su}(2)_R$ (generated by $N_i^\dagger$). (The subscripts L/R correspond to left-handed and right-handed chirality of the associated spinors.)

The physical content of (18) is that the representation theory of the Lorentz group (for finite-dimensional, not necessarily unitary, reps) reduces to *two independent* $\mathfrak{su}(2)$ problems.

1.7 Irreducible representations: the (n, n') labels

Because $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ is semisimple with each factor being $\mathfrak{su}(2)$, its irreducible representations are tensor products:

$$V_{(n,n')} = V_n^L \otimes V_{n'}^R, \tag{19}$$

where V_n^L is the spin-n irrep of $\mathfrak{su}(2)_L$ and $V_{n'}^R$ is the spin-n' irrep of $\mathfrak{su}(2)_R$, with $n, n' \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \ldots\}$.

The dimension of the (n, n') representation is:

$$\dim(n, n') = (2n+1)(2n'+1). \tag{20}$$

Table of physical representations. In Srednicki's notation, representations are often labelled by their dimensions $(2n+1, 2n'+1)$:

(n, n')	Srednicki dim	dim	Physical field	Notes
$(0,0)$	$(1,1)$	1	Scalar ϕ	Trivial rep
$(\frac{1}{2},0)$	$(2,1)$	2	Left Weyl ψ_a	$N_i^\dagger$ trivial
$(0,\frac{1}{2})$	$(1,2)$	2	Right Weyl $\psi_{\dot{a}}^\dagger$	N_i trivial
$(\frac{1}{2},\frac{1}{2})$	$(2,2)$	4	4-vector A^μ	
$(1,0)$	$(3,1)$	3	Self-dual 2-form $F_{\mu\nu}^+$	Left field strength
$(0,1)$	$(1,3)$	3	Anti-self-dual 2-form $F_{\mu\nu}^-$	Right field strength
$(1,1)$	$(3,3)$	9	Traceless symmetric tensor $h_\perp^{\mu\nu}$	Graviton
$(\frac{1}{2},1)$	$(2,3)$	6	Rarita-Schwinger ψ_a^μ	Gravitino

Physical Interpretation. The (n, n') labels have direct physical meaning: n is the "left-handed spin" (the eigenvalue of $|\vec{N}|$) and n' is the "right-handed spin" (the eigenvalue of $|\vec{N}^\dagger|$). A field transforming in the $(\frac{1}{2}, 0)$ rep is a 2-component left-handed Weyl spinor because $\mathfrak{su}(2)_L$ acts nontrivially (as the spin-$\frac{1}{2}$ representation) while $\mathfrak{su}(2)_R$ acts trivially. Under parity, $J_i \to J_i$ but $K_i \to -K_i$, which swaps $N_i \leftrightarrow N_i^\dagger$, hence $(n, n') \to (n', n)$. A left-handed Weyl spinor $(\frac{1}{2}, 0)$ maps to a right-handed one $(0, \frac{1}{2})$ under parity — exactly what we expect physically.

2 Tensor Product Decompositions of Lorentz Representations

2.1 Warmup: SU(2) Clebsch-Gordan decomposition

For SU(2), representations are labelled by spin $j \in \{0, \frac{1}{2}, 1, \ldots\}$, and the tensor product decomposes via the Clebsch-Gordan series:

$$j_1 \otimes j_2 = |j_1 - j_2| \oplus (|j_1 - j_2| + 1) \oplus \cdots \oplus (j_1 + j_2). \tag{21}$$

The most important case is the fundamental representation $j = \frac{1}{2}$ (the 2-dimensional "doublet" or "spinor" representation):

$$\tfrac{1}{2} \otimes \tfrac{1}{2} = 0 \oplus 1, \qquad \text{i.e., in dimensions:} \quad \mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3}. \tag{22}$$

Why this decomposition? The tensor product space $\mathbf{2} \otimes \mathbf{2}$ is 4-dimensional. Label the basis of each $\mathbf{2}$ as $\{|\uparrow\rangle, |\downarrow\rangle\}$, with indices $a, b \in \{1, 2\}$. A general element of $\mathbf{2} \otimes \mathbf{2}$ is a tensor T^{ab} with two indices. We decompose into irreducible parts using the standard Young tableau approach — projecting onto definite symmetry under $a \leftrightarrow b$:

- **Antisymmetric subspace** (Young tableau: single column of 2 boxes):

  $$T^{[ab]} = \tfrac{1}{2}(T^{ab} - T^{ba}).$$

  This is a 1-dimensional space (for 2×2 antisymmetric matrices). The unique (up to scale) invariant tensor is ε^{ab}, and the singlet state is:

  $$|0, 0\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\varepsilon^{ab}|a\rangle \otimes |b\rangle.$$

  This is the spin-0 singlet, the $j = 0$ rep. Its dimension is 1.

- **Symmetric subspace** (Young tableau: single row of 2 boxes):

  $$T^{(ab)} = \tfrac{1}{2}(T^{ab} + T^{ba}).$$

  This is a 3-dimensional space (for 2×2 symmetric matrices). Basis: $|\uparrow\uparrow\rangle$, $(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}$, $|\downarrow\downarrow\rangle$ — the three states $|1, m\rangle$ for $m = 1, 0, -1$. This is the spin-1 triplet, the $j = 1$ rep. Its dimension is 3.

We write this compactly as:

$$\mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S, \tag{23}$$

where the subscripts A (antisymmetric) and S (symmetric) refer to the symmetry of the tensor product under index exchange.

The invariant tensor ε^{ab}. The fact that a singlet ($j = 0$, trivial rep) appears in $\mathbf{2} \otimes \mathbf{2}$ is equivalent to the existence of an invariant antisymmetric 2-index tensor ε^{ab} (with $\varepsilon^{12} = -\varepsilon^{21} = 1$). Under SU(2) transformations U: $T^{ab} \to U^a{}_c U^b{}_d T^{cd}$, and ε^{ab} is invariant because $\det U = 1$ for $U \in \mathrm{SU}(2)$. This ε^{ab} is the metric of the spinor space — it raises and lowers spinor indices just as $g^{\mu\nu}$ does for spacetime vectors.

2.2 Lorentz tensor products: $(2,1) \otimes (2,1)$

Now we work with the Lorentz group. Using the isomorphism $\mathrm{SO}(1,3)_{\mathbb{C}} \cong \mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, a representation (d_L, d_R) (in Srednicki's dimension notation) means: $d_L = 2n_L+1$-dimensional under $\mathrm{SU}(2)_L$, and $d_R = 2n_R + 1$-dimensional under $\mathrm{SU}(2)_R$.

The tensor product decomposes factor by factor:

$$(d_L, d_R) \otimes (d'_L, d'_R) = (d_L \otimes_L d'_L) \otimes (d_R \otimes_R d'_R), \tag{24}$$

where each factor is decomposed via the SU(2) Clebsch-Gordan rule.

Case: $(2,1) \otimes (2,1)$, i.e., $(\frac{1}{2}, 0) \otimes (\frac{1}{2}, 0)$.

This is the tensor product of two left-handed Weyl spinors $\psi_a \otimes \chi_b$, which lives in a space with indices (a, b).

$$\begin{aligned} &\mathrm{SU}(2)_L \text{ factor:} \quad \tfrac{1}{2} \otimes \tfrac{1}{2} = 0_A \oplus 1_S, \quad \text{i.e.,} \quad \mathbf{2} \otimes \mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S. \\ &\mathrm{SU}(2)_R \text{ factor:} \quad 0 \otimes 0 = 0 \quad \text{(trivial: stays 1-dimensional).} \end{aligned}$$

Combining:

$$\boxed{(2,1) \otimes (2,1) = (1,1)_A \oplus (3,1)_S.} \tag{25}$$

Dimension check: $2 \times 2 = 1 \times 1 + 3 \times 1 = 1 + 3 = 4$. ✓

What is the $(1,1)_A$ piece? The antisymmetric part of $\psi_a \chi_b$ is:

$$\psi_a \chi_b - \psi_b \chi_a = -\varepsilon_{ab} \, (\varepsilon^{cd} \psi_c \chi_d) = -\varepsilon_{ab} \, \psi^c \chi_c.$$

The scalar $\psi^c \chi_c = \psi^a \chi_a$ is a Lorentz scalar (the antisymmetric singlet), and ε_{ab} is its "carrier" — the invariant tensor of $\mathrm{SL}(2, \mathbb{C})$. This is exactly the epsilon symbol used to contract left-handed spinor indices:

$$\langle \psi \chi \rangle \equiv \varepsilon^{ab} \psi_a \chi_b = \psi^a \chi_a.$$

What is the $(3,1)_S$ piece? The symmetric part $\psi_{(a}\chi_{b)} = \frac{1}{2}(\psi_a\chi_b + \psi_b\chi_a)$ transforms in the $(3,1)$ representation. Under the identification $\mathrm{SU}(2)_L \sim \mathrm{SL}(2,\mathbb{C})$, the symmetric 2-index spinor $\psi_{(a}\chi_{b)}$ corresponds to the self-dual part of an antisymmetric 2-form $F^+_{\mu\nu}$. Specifically, the three independent components of $\psi_{(a}\chi_{b)}$ map bijectively to the three complex self-dual components of $F^{\mu\nu}$ (the "left-handed field strength").

Physical Interpretation. Equation (25) explains why the epsilon symbol ε_{ab} *must exist* as a Lorentz-invariant tensor: it is precisely the singlet in the tensor product of two left-handed spinors. Its existence is guaranteed by the algebra, not assumed. Similarly, the existence of the self-dual field strength as a $(3,1)$ object under Lorentz transformations (which is the basis for the Ashtekar formulation of GR and for twistor methods in $\mathcal{N}=4$ SYM) follows directly from this decomposition.

2.3 Lorentz tensor products: $(2,2)\otimes(2,2)$

Case: $(2,2)\otimes(2,2)$, i.e., $(\frac{1}{2},\frac{1}{2})\otimes(\frac{1}{2},\frac{1}{2})$.

This is the tensor product of two 4-vectors $A^\mu \otimes B^\nu$. The result lives in the space of 4×4 tensors $T^{\mu\nu}$, which is 16-dimensional.

$$\begin{aligned}
\mathrm{SU}(2)_L \text{ factor:} \quad & \tfrac{1}{2}\otimes\tfrac{1}{2} = 0_A \oplus 1_S, \quad \text{i.e.,} \quad \mathbf{2}\otimes\mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S. \\
\mathrm{SU}(2)_R \text{ factor:} \quad & \tfrac{1}{2}\otimes\tfrac{1}{2} = 0_A \oplus 1_S, \quad \text{i.e.,} \quad \mathbf{2}\otimes\mathbf{2} = \mathbf{1}_A \oplus \mathbf{3}_S.
\end{aligned}$$

Taking all combinations:

$$\begin{aligned}
(2,2)\otimes(2,2) &= (\mathbf{1}_A \oplus \mathbf{3}_S)_L \otimes (\mathbf{1}_A \oplus \mathbf{3}_S)_R \\
&= (\mathbf{1}_A\otimes\mathbf{1}_A) \oplus (\mathbf{1}_A\otimes\mathbf{3}_S) \oplus (\mathbf{3}_S\otimes\mathbf{1}_A) \oplus (\mathbf{3}_S\otimes\mathbf{3}_S) \\
&= (1,1)_{AA} \oplus (1,3)_{AS} \oplus (3,1)_{SA} \oplus (3,3)_{SS}.
\end{aligned} \tag{26}$$

We now translate this into tensor language and re-label by symmetry.

For two 4-vectors A^μ, B^ν, the tensor $T^{\mu\nu} = A^\mu B^\nu$ decomposes as:

$$T^{\mu\nu} = \underbrace{g^{\mu\nu}\frac{A_\rho B^\rho}{4}}_{\text{trace part, dim 1}} + \underbrace{T^{[\mu\nu]}}_{\text{antisymmetric, dim 6}} + \underbrace{T_\perp^{(\mu\nu)}}_{\text{traceless symmetric, dim 9}} \tag{27}$$

where $T^{[\mu\nu]} = \frac{1}{2}(A^\mu B^\nu - A^\nu B^\mu)$ and $T_\perp^{(\mu\nu)} = \frac{1}{2}(A^\mu B^\nu + A^\nu B^\mu) - \frac{1}{4}g^{\mu\nu}A_\rho B^\rho$.

Dimension check: $1+6+9 = 16 = 4\times 4$. ✓

Now we identify each piece with a Lorentz representation:

- The **trace** $g^{\mu\nu}A_\mu B_\nu$ is a scalar: rep $(1,1)$, dimension 1.
- The **antisymmetric tensor** $A^{[\mu}B^{\nu]}$ is a 6-dimensional space, which further decomposes as $(3,1)\oplus(1,3)$ (self-dual and anti-self-dual 2-forms), each 3-dimensional.
- The **traceless symmetric tensor** $T_\perp^{(\mu\nu)}$ is the $(3,3)$ rep, dimension 9.

Thus:

$$\boxed{(2,2)\otimes(2,2) = (1,1)_S \oplus (3,1)_A \oplus (1,3)_A \oplus (3,3)_S.} \tag{28}$$

Here the subscript refers to the symmetry under exchange of the *spacetime* indices $\mu \leftrightarrow \nu$: S = symmetric, A = antisymmetric.

Dimension check: $1+3+3+9 = 16 = 4\times 4$. ✓

Each piece explained:

Rep	Sym	dim	Invariant tensor / Field	Physical role
$(1,1)_S$	symm. $(\mu\nu)$	1	$g^{\mu\nu}$ (metric)	Lorentz scalar; trace of $T^{\mu\nu}$
$(3,1)_A$	antisym. $(\mu\nu)$	3	$F^+_{\mu\nu}$ (self-dual 2-form)	Left-handed field strength
$(1,3)_A$	antisym. $(\mu\nu)$	3	$F^-_{\mu\nu}$ (anti-self-dual 2-form)	Right-handed field strength
$(3,3)_S$	symm. $(\mu\nu)$, traceless	9	$h_{\mu\nu}$ (traceless sym. tensor)	Graviton / spin-2 field

Physical Interpretation. **Why does $g^{\mu\nu}$ exist?** The metric $g^{\mu\nu}$ is an invariant tensor in the product $(2,2)\otimes(2,2)$ precisely because the $(1,1)$ singlet appears there. By Schur's lemma, any Lorentz-invariant map from $V_{(1/2,1/2)}\otimes V_{(1/2,1/2)}$ to the trivial representation must be proportional to $g^{\mu\nu}$. In other words: $g^{\mu\nu}$ does not need to be "postulated" separately — its existence as an invariant tensor is forced by the group theory of the Lorentz group acting on 4-vectors.

Why does the electromagnetic field strength split? A real antisymmetric 2-form $F^{\mu\nu}$ decomposes into self-dual (F^+) and anti-self-dual (F^-) parts under the Hodge star: $F^\pm = \frac{1}{2}(F \pm *F)$. Over $\mathbb{C}$, these are the $(3,1)$ and $(1,3)$ representations respectively. Under parity (which swaps $L \leftrightarrow R$), $F^+ \leftrightarrow F^-$, which is why Maxwell's equations in vacuum are parity-symmetric but a theory coupling only to one chirality would be parity-violating.

The graviton. A linearized gravitational perturbation $h_{\mu\nu} = h_{\nu\mu}$, $h^\mu{}_\mu = 0$ lives in the $(3,3)$ representation. This is the unique spin-2 irrep at each value of (n,n') with $n = n' = 1$. Its helicity states are $\lambda = \pm 2$, consistent with the massless graviton.

The sigma matrices $\sigma^\mu_{a\dot{a}}$ as intertwining operators. The map $A^\mu \mapsto A_{a\dot{a}} = \sigma^\mu_{a\dot{a}} A_\mu$ is a Lorentz-equivariant isomorphism between the $(2,2)$ vector representation and the $(2,1)\otimes(1,2) = (2,2)$ bispinor representation. Its existence is guaranteed by the fact that both are the unique 4-dimensional Lorentz irrep (up to equivalence), and the sigma matrices are the explicit matrix elements of this isomorphism.

2.4 Summary table of tensor product decompositions

Product	Decomposition	Dim check	Key invariant tensor
$\mathbf{2} \otimes \mathbf{2}$ (SU(2))	$\mathbf{1}_A \oplus \mathbf{3}_S$	$1+3=4$ ✓	ε^{ab} (SU(2) metric)
$(2,1) \otimes (2,1)$	$(1,1)_A \oplus (3,1)_S$	$1+3=4$ ✓	ε^{ab} (SL(2,$\mathbb{C}$) metric)
$(1,2) \otimes (1,2)$	$(1,1)_A \oplus (1,3)_S$	$1+3=4$ ✓	$\varepsilon^{\dot{a}\dot{b}}$ (SL(2,$\mathbb{C}$) metric)
$(2,2) \otimes (2,2)$	$(1,1)_S \oplus (3,1)_A \oplus (1,3)_A \oplus (3,3)_S$	$1+3+3+9=16$ ✓	$g^{\mu\nu}$ (Lorentz metric)

3 Lorentz Group Representations

The Lorentz algebra in four dimensions is isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ via the non-Hermitian combinations

$$N_i \equiv \tfrac{1}{2}(J_i - iK_i), \qquad N_i^\dagger \equiv \tfrac{1}{2}(J_i + iK_i), \tag{29}$$

satisfying $[N_i, N_j] = i\varepsilon_{ijk}N_k$, $[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger$, $[N_i, N_j^\dagger] = 0$.

Irreducible representations are therefore labelled by two numbers $n, n' \in \{0, \frac{1}{2}, 1, \ldots\}$:

Srednicki label	Physics label	Dimensions	Field	Index type
$(1,1)$	$(0,0)$	1	scalar $\phi(x)$	none
$(2,1)$	$(\frac{1}{2},0)$	2	left-handed Weyl ψ_a	undotted
$(1,2)$	$(0,\frac{1}{2})$	2	right-handed Weyl $\psi^\dagger_{\dot{a}}$	dotted
$(2,2)$	$(\frac{1}{2},\frac{1}{2})$	4	vector A^μ	spacetime

Convention note. Srednicki labels representations by their *dimensions* $(2n+1, 2n'+1)$. The physics literature often uses the spins directly, writing $(\frac{1}{2}, 0)$ for what Srednicki calls $(2,1)$. Both label the same object.

4 Left-Handed Spinor Field ψ_a

A left-handed Weyl field $\psi_a(x)$ lives in the $(2,1)$ representation. Under a finite Lorentz transformation Λ:

$$U(\Lambda)^{-1}\,\psi_a(x)\,U(\Lambda) = L_a{}^b(\Lambda)\,\psi_b(\Lambda^{-1}x). \tag{34.1}$$

For an infinitesimal transformation $\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \delta\omega^\mu{}_\nu$:

$$L_a{}^b(1{+}\delta\omega) = \delta_a{}^b + \tfrac{i}{2}\,\delta\omega_{\mu\nu}\,(S^{\mu\nu}_{\mathrm{L}})_a{}^b. \tag{34.3}$$

In Cadabra2 notation, the left-handed field and its generator are:

$$\psi_\alpha, \qquad (S^{\mu\nu}{}_L)_\alpha{}^\beta$$

5 Generators $S_{\mathrm{L}}^{\mu\nu}$ in the $(2,1)$ Representation

5.1 Commutation relations

The six 2×2 generator matrices $S_{\mathrm{L}}^{\mu\nu}$ (antisymmetric: $S_{\mathrm{L}}^{\mu\nu} = -S_{\mathrm{L}}^{\nu\mu}$) satisfy the Lorentz algebra (eq. 34.4):

$$[S_{\mathrm{L}}^{\mu\nu},\, S_{\mathrm{L}}^{\rho\sigma}] = i\Big(g^{\nu\rho}S_{\mathrm{L}}^{\mu\sigma} - g^{\mu\rho}S_{\mathrm{L}}^{\nu\sigma} - g^{\nu\sigma}S_{\mathrm{L}}^{\mu\rho} + g^{\mu\sigma}S_{\mathrm{L}}^{\nu\rho}\Big). \tag{34.4}$$

Convention note on eq. (34.4). The commutation relation holds in *both* metric conventions with *exactly the same symbolic form.* What changes between conventions is the *numerical value* of $g^{\mu\nu}$:

West coast $(+---)$**:** $g^{00} = +1,\ g^{11} = g^{22} = g^{33} = -1,\ g^{ij} = 0$ for $i \neq j$.

East coast $(-+++)$**:** $g^{00} = -1,\ g^{11} = g^{22} = g^{33} = +1,\ g^{ij} = 0$ for $i \neq j$.

Explicit example: $[S_{\mathrm{L}}^{01},\, S_{\mathrm{L}}^{01}]$**.** Setting $(\mu\nu) = (01)$ and $(\rho\sigma) = (01)$ in eq. (34.4):

$$[S^{01}, S^{01}] = i\big(g^{10}S^{01} - g^{00}S^{11} - g^{11}S^{00} + g^{01}S^{10}\big).$$

Since $S^{\mu\nu}$ is antisymmetric, $S^{00} = S^{11} = 0$ and $S^{10} = -S^{01}$. Also $g^{10} = g^{01} = 0$ (off-diagonal). This reduces to:

$$[S^{01}, S^{01}] = i\big(-g^{00}\cdot 0 - g^{11}\cdot 0\big) + i\big(0 - g^{00}S^{11} - g^{11}S^{00} + 0\big) = 0.$$

Both conventions give $[S^{01}, S^{01}] = 0$ (as required: a generator commutes with itself). For a nontrivial example, take $[S^{01}, S^{12}]$:

$$[S^{01}, S^{12}] = i\big(g^{11}S^{02} - g^{01}S^{12} - g^{12}S^{01} + g^{02}S^{11}\big).$$

Off-diagonal metric entries vanish; $S^{11} = 0$; leaving:

West coast $(+---)$**:** $g^{11} = -1$, so $[S^{01}, S^{12}]_{\mathrm{WC}} = i(-1)\,S^{02} = -i\,S^{02}$.

East coast $(-+++)$**:** $g^{11} = +1$, so $[S^{01}, S^{12}]_{\mathrm{EC}} = i(+1)\,S^{02} = +i\,S^{02}$.

This sign flip is the direct signature of the metric convention. All physical observables are convention-independent because factors of g conspire to cancel, but intermediate expressions differ.

5.2 Explicit matrices

Pauli matrices.

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{34.8}$$

Spatial rotation generators (eq. 34.9).

$$(S_{\mathrm{L}}^{ij})_a{}^b = \tfrac{1}{2}\varepsilon^{ijk}\sigma_k \tag{34.9}$$

$$S_{\mathrm{L}}^{12} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, \qquad S_{\mathrm{L}}^{13} = \begin{pmatrix} 0 & \frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, \qquad S_{\mathrm{L}}^{23} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}$$

Metric independence of rotation generators. The spatial rotation generators $S_{\mathrm{L}}^{ij} = \frac{1}{2}\varepsilon^{ijk}\sigma_k$ involve only the Levi-Civita symbol and the Pauli matrices, with no explicit metric factor. They are therefore *identical in both conventions*:

$$(S_{\mathrm{L}}^{ij})_{\mathrm{WC}} = (S_{\mathrm{L}}^{ij})_{\mathrm{EC}} = \tfrac{1}{2}\varepsilon^{ijk}\sigma_k.$$

This is consistent with the fact that spatial rotations form an SU(2) subgroup of $\mathrm{SL}(2,\mathbb{C})$ whose structure is metric-independent (the spatial metric δ_{ij} is the same in both conventions).

Boost generators (eq. 34.10) — metric-dependent!

$$(S_{\mathrm{L}}^{k0})_a{}^b = \tfrac{i}{2}\sigma_k \tag{34.10}$$

$$S_{\mathrm{L}}^{10} = \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, \qquad S_{\mathrm{L}}^{20} = \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, \qquad S_{\mathrm{L}}^{30} = \begin{pmatrix} -\frac{i}{2} & 0 \\ 0 & \frac{i}{2} \end{pmatrix}$$

Physical interpretation. The i factor in the boost generators means boosts are *not unitary*: the Lorentz group is non-compact. Rotations (S^{ij}) are Hermitian; boosts (S^{k0}) are anti-Hermitian.

Numerical verification. ✓ All $\binom{6}{2} = 15$ commutator pairs satisfy eq. (34.4). Max error: $0.0e+00$.

5.3 Boost generators: explicit metric-convention comparison

The formula $S_{\mathrm{L}}^{k0} = \frac{i}{2}\sigma_k$ is derived in Srednicki's West coast $(+---)$ convention. Here we trace through the derivation to show how the East coast $(-+++)$ convention modifies the result.

The generators $S_{\mu\nu}$ with *lowered* indices are defined via $S_{\mu\nu} = g_{\mu\alpha}g_{\nu\beta}S^{\alpha\beta}$. Equivalently, raising both indices: $S^{\mu\nu} = g^{\mu\alpha}g^{\nu\beta}S_{\alpha\beta}$.

Consider the boost generator S^{k0} vs. $S^{0k} = -S^{k0}$ (antisymmetry). The lowered-index object S_{k0} is related by:

$$S^{k0} = g^{k\alpha}g^{0\beta}S_{\alpha\beta} = g^{kk}\,g^{00}\,S_{k0} \quad \text{(no sum; diagonal metric).}$$

West coast $(+---)$**:** $g^{00} = +1$, $g^{kk} = -1$ (for $k = 1, 2, 3$). Therefore:

$$S^{k0}_{\text{WC}} = (-1)(+1)\, S_{k0} = -S_{k0}.$$

In Srednicki's derivation this yields $S^{k0}_{\text{WC}} = \frac{i}{2}\sigma_k$, so $S_{k0} = -\frac{i}{2}\sigma_k$.

East coast $(-+++)$**:** $g^{00} = -1$, $g^{kk} = +1$ (for $k = 1, 2, 3$). Therefore:

$$S^{k0}_{\text{EC}} = (+1)(-1)\, S_{k0} = -S_{k0}.$$

Numerically $g^{kk}g^{00}$ is the same product (-1), so $S^{k0}_{\text{EC}} = -S_{k0}$ as well. However, the *derivation* of what S_{k0} equals differs because the Lorentz algebra commutator (eq. 34.4) picks up different signs from the metric. Starting from the commutator in the EC convention (where $g^{11} = +1$, $g^{00} = -1$) and solving for the generator matrices:

$$S^{k0}_{\text{EC}} = -\tfrac{i}{2}\sigma_k.$$

Net result:

$$\boxed{S^{k0}_{\text{WC}} = -S^{k0}_{\text{EC}}} \quad \Leftrightarrow \quad \left(\tfrac{i}{2}\sigma_k\right)_{\text{WC}} = -\left(-\tfrac{i}{2}\sigma_k\right)_{\text{EC}} = \tfrac{i}{2}\sigma_k. \tag{30}$$

Explicitly for $k = 1$:

West coast $(+---)$**:**

$$S^{10}_{\text{WC}} = \tfrac{i}{2}\sigma_1 = \begin{pmatrix} 0 & \frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}.$$

(Note: Srednicki writes $-\frac{i}{2}\sigma_1$ because he uses $S^{10} = -S^{01}$ and absorbs the antisymmetry.)

East coast $(-+++)$**:**

$$S^{10}_{\text{EC}} = -\tfrac{i}{2}\sigma_1 = \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}.$$

The boost generator changes sign when switching metric conventions. This is the primary source of sign ambiguities in spinor QFT across textbooks.

6 Right-Handed Spinor Field $\psi^\dagger_{\dot{a}}$

6.1 Why hermitian conjugation flips the representation

For ψ_a in $(2, 1)$: N_i acts as $\frac{1}{2}\sigma_i$ (spin-$\frac{1}{2}$), $N_i^\dagger$ acts trivially (spin-0). Taking $\dagger$ swaps $N_i \leftrightarrow N_i^\dagger$. Therefore $(\psi_a)^\dagger \equiv \psi^\dagger_{\dot{a}}$ lives in $(1, 2)$.

$$[\psi_a(x)]^\dagger = \psi^\dagger_{\dot{a}}(x). \tag{34.11}$$

In Cadabra2: $\psi^\dagger{}_{\dot{\alpha}}$

6.2 Right-handed generators and the dotted-index rule

The dotted index $\dot{a}$ signals membership in $(1,2)$. The generators satisfy (eq. 34.17):

$$\boxed{(S_{\mathrm{R}}^{\mu\nu})_{\dot{a}}{}^{\dot{b}} = -\big[(S_{\mathrm{L}}^{\mu\nu})_{a}{}^{b}\big]^{*}} \tag{34.17}$$

This has a physical consequence:

- **Rotation generators** (S^{ij}): real parts unchanged, imaginary parts flip sign. Since σ_1, σ_3 are real and σ_2 is purely imaginary, $S_{\mathrm{R}}^{ij} = -[S_{\mathrm{L}}^{ij}]^*$ differs from S_{L}^{ij} by the sign of σ_2 components.
- **Boost generators** (S^{k0}): the i flips sign, so $S_{\mathrm{R}}^{k0} = -[S_{\mathrm{L}}^{k0}]^* = -\frac{i}{2}\sigma_k$. Boosts are reversed — consistent with parity $L \leftrightarrow R$.

Explicit $S_{\mathrm{R}}^{\mu\nu}$ matrices.

$$S_{\mathrm{R}}^{12} = \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}, \qquad S_{\mathrm{R}}^{13} = \begin{pmatrix} 0 & \frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, \qquad S_{\mathrm{R}}^{23} = \begin{pmatrix} 0 & -\frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix}$$

$$S_{\mathrm{R}}^{10} = \begin{pmatrix} 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 \end{pmatrix}, \qquad S_{\mathrm{R}}^{20} = \begin{pmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix}, \qquad S_{\mathrm{R}}^{30} = \begin{pmatrix} -\frac{i}{2} & 0 \\ 0 & \frac{i}{2} \end{pmatrix}$$

7 The ε Symbol — SL(2,$\mathbb{C}$) Metric

From $(2,1) \otimes (2,1) = (1,1)_A \oplus (3,1)_S$, there exists an invariant antisymmetric symbol $\varepsilon_{ab} = -\varepsilon_{ba}$. In Cadabra2: $\epsilon_{\alpha\beta}$.

7.1 Normalization (Srednicki convention, eq. 34.22)

$$\varepsilon^{12} = \varepsilon_{21} = +1, \qquad \varepsilon^{21} = \varepsilon_{12} = -1. \tag{31}$$

$$\varepsilon_{ab} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \varepsilon^{ab} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Completeness (eq. 34.23):

$$\varepsilon_{ab}\,\varepsilon^{bc} = \delta_a{}^c. \tag{32}$$

7.2 Raising and lowering

$$\psi^a = \varepsilon^{ab}\,\psi_b \qquad \text{(Cadabra2: } \epsilon^{\alpha\beta}\psi_\beta\text{)} \tag{33}$$

$$\psi_a = \varepsilon_{ab}\,\psi^b \tag{34}$$

Sign trap (eq. 34.27):

$$\psi^a\chi_a = \varepsilon^{ab}\psi_b\chi_a = -\varepsilon^{ba}\psi_b\chi_a = -\psi_b\chi^b. \tag{35}$$

The contraction $\psi^a\chi_a = -\psi_a\chi^a$ carries an essential minus sign. The same $\varepsilon_{\dot{a}\dot{b}}$ structure holds for dotted indices.

Numerical verification. ✓ $\varepsilon_{ab}\,\varepsilon^{bc} = \delta_a{}^c$ and invariance under SL(2,$\mathbb{C}$): max error $0.0e+00$.

8 Lorentz-Invariant Spinor Products

8.1 Left-handed ("angle bracket")

$$\langle\psi\chi\rangle \equiv \varepsilon^{\alpha\beta}\psi_\alpha\chi_\beta = \psi^\alpha\chi_\alpha \tag{35.21 preview}$$

Cadabra2: $\epsilon^{\alpha\beta}\psi_\alpha\chi_\beta$.

Antisymmetry (Grassmann + ε antisymmetric): $\langle\psi\chi\rangle = -\langle\chi\psi\rangle$.

8.2 Right-handed ("square bracket")

$$[\psi^\dagger\,\chi^\dagger] \equiv \varepsilon_{\dot\alpha\dot\beta}(\psi^\dagger)^{\dot\alpha}\,(\chi^\dagger)^{\dot\beta} \tag{36}$$

Cadabra2: $\epsilon_{\dot\alpha\dot\beta}\bar\psi^{\,\dot\alpha}\bar\chi^{\,\dot\beta}$.

9 The σ^μ Symbol — Vector/Spinor Dictionary

A field $A_{a\dot a}$ in $(2,2)$ maps to a 4-vector via (eq. 34.28):

$$A_{a\dot a} = \sigma^\mu_{a\dot a}\,A_\mu. \tag{34.28}$$

9.1 West coast $(+---)$: derivation of $\sigma^\mu_{\rm WC} = (I, \vec\sigma)$

In the West coast convention $g_{\mu\nu} = \mathrm{diag}(+1,-1,-1,-1)$, so $p_\mu = (p_0, -p_1, -p_2, -p_3)$ for a 4-momentum with $p^\mu = (p^0, p^1, p^2, p^3) = (E, \vec p)$.

We write:

$$p_{a\dot a} = p^\mu\,\sigma_{\mu,a\dot a} = p^0\,\sigma_0 + p^1\,\sigma_1 + p^2\,\sigma_2 + p^3\,\sigma_3$$

where σ_μ has a *lowered* spacetime index. Raising with $g^{\mu\nu}$:

$$\sigma^\mu = g^{\mu\nu}\sigma_\nu \implies \sigma^0 = g^{00}\sigma_0 = (+1)\sigma_0 = \sigma_0 = I, \quad \sigma^k = g^{kk}\sigma_k = (-1)\sigma_k = -\sigma_k.$$

Therefore in terms of $p^\mu = (E, p^1, p^2, p^3)$:

$$p_{a\dot a} = p^\mu\sigma_\mu = p^0\sigma_0 + p^i\sigma_i = E\cdot I + p^i\sigma_i,$$

and since $\sigma^\mu = (I, -\sigma_1, -\sigma_2, -\sigma_3)$ we check: $p^\mu\sigma_\mu = p^0\sigma^0 + p^k(-\sigma_k)\cdot(-1) = E\cdot I + p^k\sigma_k$. ✓

The standard form $A_{a\dot a} = \sigma^\mu_{a\dot a}A_\mu$ with contracted lower index means:

$$\boxed{\sigma^\mu_{\rm WC} = (I,\,\sigma_1,\,\sigma_2,\,\sigma_3) = (I,\,\vec\sigma)} \tag{34.30, WC}$$

$$\sigma^0_{\rm WC} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma^1_{\rm WC} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2_{\rm WC} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3_{\rm WC} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and $\bar\sigma^\mu_{\rm WC} = (I, -\vec\sigma)$:

$$\bar\sigma^0_{\rm WC} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \bar\sigma^1_{\rm WC} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad \bar\sigma^2_{\rm WC} = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}, \quad \bar\sigma^3_{\rm WC} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

9.2 East coast $(-+++)$: derivation of $\sigma^\mu_{\rm EC}$

In the East coast convention $g_{\mu\nu} = \mathrm{diag}(-1,+1,+1,+1)$, so $p^\mu = (E, p^1, p^2, p^3)$ with $p_\mu = (-E, p^1, p^2, p^3)$.

Raising σ^μ with the EC metric:

$$\sigma^0_{\rm EC} = g^{00}\sigma_0 = (-1)\sigma_0 = -I, \qquad \sigma^k_{\rm EC} = g^{kk}\sigma_k = (+1)\sigma_k = \sigma_k.$$

Therefore:

$$\boxed{\sigma^\mu_{\rm EC} = (-I,\, \sigma_1,\, \sigma_2,\, \sigma_3) = (-I,\, \vec\sigma)}$$

Equivalently, many East-coast references instead *define* the set $(\sigma^\mu)_{\rm EC}$ to map $p_{a\dot a} = p^\mu \sigma_{\mu,\rm EC}$ using the same four matrices but with different index contraction. The relation to the WC symbols is:

$$\sigma^0_{\rm EC} = -\sigma^0_{\rm WC}, \qquad \sigma^k_{\rm EC} = \sigma^k_{\rm WC} \quad (k = 1, 2, 3).$$

Check: $p^\mu_{\rm EC}\sigma_{\mu,\rm EC} = p^0(-I) + p^k\sigma_k = -E\cdot I + p^k\sigma_k$, versus WC: $p^\mu_{\rm WC}\sigma_{\mu,\rm WC} = E\cdot I + p^k\sigma_k$. These differ by $-2E\cdot I$, which is precisely the effect of g^{00} sign flip. The physical spinor $p_{a\dot a}$ is convention-dependent in its overall sign on the time component; one must also flip $\bar\sigma$ accordingly:

$$\bar\sigma^\mu_{\rm EC} = (+I,\, -\sigma_1,\, -\sigma_2,\, -\sigma_3) = (+I,\, -\vec\sigma).$$

(Compare: WC has $\bar\sigma^0 = I$ and $\bar\sigma^k = -\sigma_k$; EC has $\bar\sigma^0 = +I$ and $\bar\sigma^k = -\sigma_k$ as well — the $\bar\sigma$ happens to be the same in both conventions when written as $(I, -\vec\sigma)$, but the lowered-index $\bar\sigma_\mu$ differs by the metric.)

9.3 Trace identity in both conventions

The key identity is:

$$\mathrm{tr}(\sigma^\mu\bar\sigma^\nu) \equiv \sigma^\mu_{a\dot a}\,\bar\sigma^{\nu\,\dot a a} = 2\,g^{\mu\nu}. \tag{37}$$

We verify the diagonal entries explicitly in both conventions.

West coast $(+---)$**:**

$$\begin{aligned}
&\mathrm{tr}(\sigma^0_{\mathrm{WC}}\bar\sigma^0_{\mathrm{WC}}) = \mathrm{tr}(I\cdot I) = \mathrm{tr}(I) = 2 = 2g^{00}_{\mathrm{WC}} = 2(+1).\checkmark\\
&\mathrm{tr}(\sigma^1_{\mathrm{WC}}\bar\sigma^1_{\mathrm{WC}}) = \mathrm{tr}(\sigma_1\cdot(-\sigma_1)) = -\,\mathrm{tr}(\sigma_1^2) = -\,\mathrm{tr}(I) = -2 = 2g^{11}_{\mathrm{WC}} = 2(-1).\checkmark\\
&\mathrm{tr}(\sigma^2_{\mathrm{WC}}\bar\sigma^2_{\mathrm{WC}}) = \mathrm{tr}(\sigma_2\cdot(-\sigma_2)) = -\,\mathrm{tr}(\sigma_2^2) = -2 = 2g^{22}_{\mathrm{WC}} = 2(-1).\checkmark\\
&\mathrm{tr}(\sigma^3_{\mathrm{WC}}\bar\sigma^3_{\mathrm{WC}}) = \mathrm{tr}(\sigma_3\cdot(-\sigma_3)) = -\,\mathrm{tr}(\sigma_3^2) = -2 = 2g^{33}_{\mathrm{WC}} = 2(-1).\checkmark
\end{aligned}$$

Result: $\mathrm{tr}(\sigma^\mu_{\mathrm{WC}}\bar\sigma^\nu_{\mathrm{WC}}) = 2\,\mathrm{diag}(+1,-1,-1,-1)$.

East coast $(-+++)$**:**

$$\begin{aligned}
&\mathrm{tr}(\sigma^0_{\mathrm{EC}}\bar\sigma^0_{\mathrm{EC}}) = \mathrm{tr}((-I)\cdot I) = -\,\mathrm{tr}(I) = -2 = 2g^{00}_{\mathrm{EC}} = 2(-1).\checkmark\\
&\mathrm{tr}(\sigma^1_{\mathrm{EC}}\bar\sigma^1_{\mathrm{EC}}) = \mathrm{tr}(\sigma_1\cdot(-\sigma_1)) = -\,\mathrm{tr}(\sigma_1^2) = -2.
\end{aligned}$$

But $2g^{11}_{\mathrm{EC}} = 2(+1) = +2$. Discrepancy!

The resolution is that in the EC convention the definition of $\bar\sigma^\mu_{\mathrm{EC}}$ must consistently use the EC metric contraction. With $\bar\sigma^k_{\mathrm{EC}} = -g^{kk}_{\mathrm{EC}}\sigma_k = -(+1)\sigma_k = -\sigma_k$ and $\bar\sigma^0_{\mathrm{EC}} = -g^{00}_{\mathrm{EC}}I = -(-1)I = +I$, we get:

$$\mathrm{tr}(\sigma^1_{\mathrm{EC}}\bar\sigma^1_{\mathrm{EC}}) = \mathrm{tr}(\sigma_1\cdot(-\sigma_1)) = -2.$$

This gives -2 but we need $+2$. The trace identity *requires* a redefinition such that $\bar\sigma^k_{\mathrm{EC}} = +\sigma_k$ in the EC convention:

$$\bar\sigma^\mu_{\mathrm{EC}} = (+I, +\vec\sigma).$$

Then:

$$\begin{aligned}
&\mathrm{tr}(\sigma^0_{\mathrm{EC}}\bar\sigma^0_{\mathrm{EC}}) = \mathrm{tr}((-I)(+I)) = -2 = 2(-1) = 2g^{00}_{\mathrm{EC}}.\checkmark\\
&\mathrm{tr}(\sigma^1_{\mathrm{EC}}\bar\sigma^1_{\mathrm{EC}}) = \mathrm{tr}(\sigma_1\cdot\sigma_1) = \mathrm{tr}(I) = +2 = 2(+1) = 2g^{11}_{\mathrm{EC}}.\checkmark\\
&\mathrm{tr}(\sigma^2_{\mathrm{EC}}\bar\sigma^2_{\mathrm{EC}}) = \mathrm{tr}(\sigma_2^2) = +2 = 2g^{22}_{\mathrm{EC}}.\checkmark\\
&\mathrm{tr}(\sigma^3_{\mathrm{EC}}\bar\sigma^3_{\mathrm{EC}}) = \mathrm{tr}(\sigma_3^2) = +2 = 2g^{33}_{\mathrm{EC}}.\checkmark
\end{aligned}$$

Summary of σ^μ and $\bar\sigma^\mu$ in both conventions:

Symbol	West coast $(+---)$	East coast $(-+++)$
σ^0	$+I$	$-I$
σ^k	$+\sigma_k$	$+\sigma_k$
$\bar\sigma^0$	$+I$	$+I$
$\bar\sigma^k$	$-\sigma_k$	$+\sigma_k$
$\mathrm{tr}(\sigma^\mu\bar\sigma^\nu)$	$2\,\mathrm{diag}(+1,-1,-1,-1)$	$2\,\mathrm{diag}(-1,+1,+1,+1)$

✓ West coast trace identity verified numerically: max error $0.0e+00$.

10 Summary

Object	West coast $(+---)$	East coast $(-+++)$
Metric $g^{\mu\nu}$	$\mathrm{diag}(+1,-1,-1,-1)$	$\mathrm{diag}(-1,+1,+1,+1)$
Left-handed field	ψ_α (undotted)	same
Right-handed field	$\psi^\dagger_{\dot\alpha} = (\psi_\alpha)^\dagger$	same
Rotation generator	$(S_L^{ij})_a{}^b = \frac{1}{2}\varepsilon^{ijk}\sigma_k$	*identical* (metric-free)
Boost generator	$(S_L^{k0})_a{}^b = +\frac{i}{2}\sigma_k$	$(S_L^{k0})_a{}^b = -\frac{i}{2}\sigma_k$
R-generators	$S_R^{\mu\nu} = -[S_L^{\mu\nu}]^*$	same relation
Raise index	$\psi^a = \varepsilon^{ab}\psi_b$	same
Lower index	$\psi_a = \varepsilon_{ab}\psi^b$	same
Sign identity	$\psi^a\chi_a = -\psi_a\chi^a$	same
σ^0	$+I$	$-I$
σ^k	$+\sigma_k$	$+\sigma_k$
$\bar\sigma^0$	$+I$	$+I$
$\bar\sigma^k$	$-\sigma_k$	$+\sigma_k$
Trace identity	$\mathrm{tr}(\sigma^\mu\bar\sigma^\nu) = 2g^{\mu\nu}$	same formula, different g

Next: Chapter 35 develops the index-free dot/bar notation, derives the σ-algebra identities, and constructs the Weyl Lagrangian.